

Pythagoras' Theorem – Answer Sheet

WorksheetHub AI
Mathematics · Grade 7

Unit 3 · Geometry

Hard

Pythagoras theorem

A. Find the hypotenuse

- 1 $36 + 64 = 100$; $c = 10$ cm
- 2 $25 + 144 = 169$; $c = 13$ cm
- 3 $36 + 49 = 85$; $c = \sqrt{85} = 9.2$ cm

B. Find a shorter side

- 4 $169 - 25 = 144$; leg = 12 cm
- 5 $100 - 36 = 64$; leg = 8 cm
- 6 $81 - 16 = 65$; leg = $\sqrt{65} = 8.1$ cm

C. Is it a right-angled triangle?

- 7 $49 + 576 = 625 = 25^2 \rightarrow$ YES, right-angled
- 8 $36 + 64 = 100$, but $11^2 = 121 \rightarrow$ NO

D. Real-life problems

- 9 $25 - 9 = 16$; $h = 4$ m
- 10 $576 + 49 = 625$; diagonal = 25 cm
- 11 $6400 + 3600 = 10000$; distance = 100 m

★ Challenge

- 12 Across = 3, up = 4; distance = $\sqrt{9 + 16} = 5$
- 13 $2s^2 = 100$; $s^2 = 50$; $s = \sqrt{50} = 7.1$ cm